DIFFERENTIALS OF A MORPHISM LOCALLY OF FINITE TYPE OF SPACES FORM A FINITE TYPE MODULE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1249-1255 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 1257-1262, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The companion finite-presentation upgrade is collected separately as L16051.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Etale-locality of the finite-type property for the module of differentials (lemma-localize-differentials) and the schemes case (Morphisms, lemma-finite-type-differentials).

Source reference. Original lemma 04CU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 981–996. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01V2: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1262–1262. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. If f is locally of finite type, then $\Omega_{X/Y}$ is a finite type $\mathcal{O}_X$-module.*

Proof. Follows from the schemes version, see Morphisms, Lemma 01V2 and étale localization, see Lemma 04CU. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.